The Three-Prime Running Least Common Multiple

Exact structure and the denominator-dependent local-window frontier

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July 2026

Abstract

Let p, q, r be pairwise distinct primes and let $\mathrm{L}(x)$ denote the least common multiple of all $\{p, q, r\}$-smooth numbers not exceeding x. We prove in Lean that

$$\mathrm{L}(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor} \qquad (x \geq 1),$$

so the running least common multiple is the product of the three maximal pure prime powers below the cutoff, and is at most x^3. Consequently it is constant on each cell where the three integer logarithms are constant; when exactly one logarithm advances by one it is multiplied by exactly the corresponding prime; and the first n positive powers in the three channels, together with the common origin 1, give exactly $3n+1$ distinct jump values. We prove an exact finite normal form in which a rectangular lattice sum of reciprocal running values is grouped by its genuine height, with each coefficient the cardinality of the corresponding fibre, and a uniform bound $9\,\#\mathcal{S} \leq (j+3)^2$ on the cardinality of a smooth exponent shell whose sorted height coordinates sum to j. At $\{2, 3, 5\}$ we exhibit the smallest exact obstruction to separating the kernel as a product, the two sides being $1/60$ and $1/12$.

Problem #269 is open and nothing here decides it. We prove no irrationality or transcendence statement in any three-prime case. What is new at the frontier is an exact consumer. For a positive reduced denominator B coprime to 30, a denominator-dependent cofinal local-window residue escape rules out every positive integral carry obeying the matching bound. Lean checks the complete implication, including the zero-residue convention and all modulus edge cases. For the actual dyadic compression, Lean proves that each open block $(2^a, 2^{a+1})$ contains at most one pure 3-power and at most one pure 5-power. Hence its block radix belongs to the exact alphabet $\{2, 6, 10, 30\}$, rather than merely lying in a coarse bounded interval. An integer-only checker constructs the corresponding block digits, reproduces three small certificates, and finds an escaping window of length at most 18 for each of 106,666 tested denominator/start pairs. This is finite experimental evidence, not a cofinal theorem. What remains unproved is the producer: bind the actual block digit and denominator-cleared carry to the formal recurrence, then prove that these four-symbol windows escape cofinally.

Authorship and AI use. The prose in this version was generated by large-language-model agents under Will Cook's direction. Cook set the objectives and acceptance criteria, reviewed and authorised the manuscript, and is responsible for its contents. The agents are production tools, not authors. See *Statements and declarations*.

1 The problem

Let P be a finite set of primes with $|P| \geq 2$, and let $a_1 < a_2 < \cdots$ enumerate the positive integers all of whose prime factors lie in P. Erdős Problem #269 asks whether

$$\sum_{n\geq 1} \frac{1}{[a_1, \dots, a_n]}$$

is irrational, where $[a_1, \dots, a_n]$ is the least common multiple [1, p. 65][2, p. 106]. Numbering and status follow Bloom's catalogue [4]. The problem is open for every finite P with $|P| \geq 2$.

Three things are known and fix the shape of the question. The restriction $|P| \geq 2$ is necessary: for $P = \{p\}$ the enumeration is $a_n = p^{n-1}$, so $[a_1, \dots, a_n] = p^{n-1}$ and the sum is $p/(p-1)$, which is rational. For infinite P the sum is always irrational, which Erdős calls a simple exercise [2, p. 106]. And in a letter of 1 January 1973 he recorded that he could prove irrationality once duplicate summands are removed [3].

That last remark is the one this note is closest to. The running least common multiple is not injective in n: it is constant along stretches of the enumeration and changes only at certain points. Removing duplicate summands means summing over the distinct values rather than over n. Sections 3 and 4 make that reindexing exact, in the case $|P| = 3$: they identify where the value is constant, by exactly what factor it changes when it changes, and what multiplicity each distinct value carries. They do not recover the irrationality proof Erdős reported for the de-duplicated sum, and they say nothing about the original sum.

Throughout, p, q, r are pairwise distinct primes. Call n *smooth* when $n = p^i q^j r^k$ for some $i, j, k \geq 0$; this is the smooth lattice value. For $x \geq 1$ write

$$\mathrm{L}(x) = \mathrm{lcm}\{\, n \leq x : \ n \text{ smooth} \,\}, \qquad \mathrm{H}(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor},$$

the running least common multiple and the pure-power height, where $\lfloor \log_b x \rfloor$ is the integer logarithm, the largest e with $b^e \leq x$. Since $a_1, \dots, a_n$ are exactly the smooth numbers up to a_n, we have $[a_1, \dots, a_n] = \mathrm{L}(a_n)$, and the summands of the problem are the reciprocals of L along the enumeration. The reciprocal of the height at a smooth point is the lattice kernel

$$\mathrm{K}(i, j, k) = \frac{1}{\mathrm{H}(p^i q^j r^k)}.$$

Scope of this note, and relation to prior work. We treat $|P| = 3$ throughout, writing $P = \{p, q, r\}$, and the smallest instance is $\{2, 3, 5\}$. Nothing here is specific to three generators for its own sake; three is where the coordinates stop separating, which Theorem 6.1 makes exact at the smallest instance. The statement of Problem #269 has been formalised before, as a conjecture with an unfilled proof, in the *Formal Conjectures* collection [5]. That is a formal statement of the question. The declarations described below are propositions about the objects the question is posed over. In relation to the historical record, the result to foreground is not the elementary running-lcm formula by itself but the checked denominator-dependent local-window consumer: it isolates a precise producer whose proof would turn the four-symbol radix structure into an irrationality contradiction. No claim of priority is made for any of these declarations.

Statement	Status	Treatment here
Irrationality in any three-prime case	Open	Not proved.
$\mathrm{L}(x) = \mathrm{H}(x)$	Proved here	Theorem 2.1; needs the primes distinct.
$\mathrm{H}(x) \le x^3$	Proved here	Proposition 2.2.
Constancy on logarithmic cells	Proved here	Theorem 3.1.
Single-coordinate jump ratios	Proved here	Theorem 3.2.
Exactly $3n + 1$ jump values	Proved here	Theorem 3.3.
Height-fibre normal form	Proved here; finite	Theorem 4.1.
Infinite limit of that expansion	Not proved	Section 4.
Shell multiplicity $9\,\#\mathcal{S} \le (j+3)^2$	Proved here	Theorem 5.3.
Kernel is not separable at $\{2, 3, 5\}$	Proved here	Theorem 6.1.
Actual dyadic block radix is in $\{2, 6, 10, 30\}$	Proved here	Theorem 7.1.
Canonical least-positive-residue arithmetic	Proved here	Theorem 7.2.
Denominator-dependent local-window consumer	Proved here	Theorem 7.3.
Dyadic-window scan for $B \le 1000$	Exact finite computation	106,666 pairs; no failures; not a theorem.
Local-window escape for the actual $\{2, 3, 5\}$ word	Open	Problem 8.1.

Status of everything below. *Status.* The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Reading map. Section 2 identifies the running value. Sections 3 and 4 develop its jump structure and the finite normal form, Section 5 bounds fibre multiplicity, and Section 6 records the obstruction to separating the kernel. Section 7 identifies the exact dyadic radix, exercises the finite consumer, and isolates the remaining cofinal producer. Every unproved hypothesis is marked as such. Linked phrases open the corresponding Lean declaration at the pinned source revision 773d2a6ec564.

Keywords. irrationality; least common multiple; smooth numbers; lattice sums; Lean 4.
MSC 2020. 11J72 (primary); 11A05, 11N25, 68V20 (secondary).

2 The running value is a product of pure powers

The smooth numbers up to x are indexed by the exponent triples (i, j, k) with $i \le \lfloor \log_p x \rfloor$, $j \le \lfloor \log_q x \rfloor$, $k \le \lfloor \log_r x \rfloor$ and $p^i q^j r^k \le x$: the coordinate bounds make the index set finite, and the last condition is the actual constraint. This is the smooth prefix index set. Both conditions matter: the coordinate box is strictly larger than the prefix, since a product of three large pure powers can exceed x while each factor does not.

Theorem 2.1 (the running least common multiple)**.** *Let p, q, r be pairwise distinct primes and $x \ge 1$. Then $\mathrm{L}(x) = \mathrm{H}(x)$.*

Proof. For divisibility in one direction, every smooth $n \le x$ has exponents bounded by the corresponding integer logarithms, so $n \mid \mathrm{H}(x)$, and hence $\mathrm{L}(x) \mid \mathrm{H}(x)$. For the other, the three pure powers $p^{\lfloor \log_p x \rfloor}$, $q^{\lfloor \log_q x \rfloor}$ and $r^{\lfloor \log_r x \rfloor}$ are themselves smooth numbers not exceeding x, so each divides $\mathrm{L}(x)$. Distinct primes have coprime powers, so their product divides $\mathrm{L}(x)$ as well. The two divisibilities give equality. □

Formalised as the running-lcm identity, from the divisibility into the height and the three membership statements for the pure powers, namely the first, second, and third pure-power memberships.

The hypothesis that the primes are pairwise distinct is used exactly once, in the coprimality step, and it is not decorative: without it the three pure powers need not have coprime orders and their product need not divide the least common multiple. The identity is what makes every later statement about L computable from three integer logarithms.

Proposition 2.2. *For every $x \ge 1$, $\mathrm{H}(x) \le x^3$.*

Proof. Each of the three factors is a power of its base not exceeding x. □

Formalised as the cubic majorant, which needs no primality. The exponent 3 is the number of generating primes, and the bound is the reason the reciprocal kernel is comparable to x^{-3} rather than to x^{-1}.

3 Cells, jumps, and their count

Say that x and y lie in the same *logarithmic cell* when $\lfloor \log_b x \rfloor = \lfloor \log_b y \rfloor$ for each of $b = p, q, r$: the cell relation.

Theorem 3.1 (constancy on cells)**.** *If $x, y \ge 1$ lie in the same logarithmic cell, then $\mathrm{L}(x) = \mathrm{L}(y)$. The same holds for the kernel at two smooth points whose values lie in one cell.*

Proof. Immediate from Theorem 2.1, since H depends on x only through the three integer logarithms. □

Formalised as the cell constancy of the running value, the cell constancy of the height, and the cell constancy of the kernel. The running value therefore changes only when one of the three logarithms changes, and the next theorem says by exactly how much.

Theorem 3.2 (single-coordinate jump ratios)**.** *Let* $x, y \geq 1$*. If* $\lfloor \log_p y \rfloor = \lfloor \log_p x \rfloor + 1$ *while the other two logarithms agree, then* $\mathrm{L}(y) = p\,\mathrm{L}(x)$*; and similarly with* q *or* r *in place of* p*.*

Proof. By Theorem 2.1 both sides are heights, and one factor of the height gains one in its exponent while the others are unchanged. □

Formalised as the first, second, and third coordinate steps, over the corresponding statements for the height alone, which need no primality: the first height step, the second, and the third.

The jump values are therefore the pure prime powers, and they do not collide across channels.

Theorem 3.3 (jump count)**.** *Let* $n \geq 0$*. The set of the first* n *positive powers of* p*, of* q*, and of* r *has exactly* $3n$ *elements, and adjoining the common origin* 1 *gives exactly* $3n + 1$*.*

Proof. Within one channel the powers $b, b^2, \ldots, b^n$ are distinct because $b \geq 2$. Across two channels a common value would be a positive power of two distinct primes, which unique factorisation forbids. Finally 1 is not a positive power of any prime. □

Formalised as the positive jump count and the jump count with the origin, over the channel cardinality, the channel disjointness, and the exclusion of the origin; the channels themselves are the positive power sets. The exponent 0 is omitted from each channel because it is the shared initial value, which is why the origin is counted once rather than three times.

4 The height-fibre normal form

Grouping a lattice sum by the value of the running least common multiple turns it into a sum over heights whose coefficients are multiplicities. We prove this exactly, on a finite rectangular box.

Write $B(h_p, h_q, h_r)$ for the box of exponent triples with $i \leq h_p$, $j \leq h_q$, $k \leq h_r$, the exponent box, and for a height H write $F(H)$ for the set of points of the box whose running height equals H, the height fibre.

Theorem 4.1 (finite normal form)**.** *For every box,*

$$\sum_{(i,j,k)\in B} \mathrm{K}(i,j,k) = \sum_{H} \frac{\#F(H)}{H},$$

the outer sum ranging over the heights actually attained on B*.*

Proof. Partition B into the fibres of the height map. On $F(H)$ every summand is $1/H$ by definition of the kernel, so the fibre contributes $\#F(H)/H$. □

Formalised as the height-fibre normal form, over the fibre sum, with the height of a lattice point given by the point height.

Together with Theorem 3.1 this is the finite core of the ordered prime-power jump expansion: the value is constant on cells, the cells are indexed by heights, and the coefficient of a height is the number of lattice points it collects. The passage to the infinite sum, and the explicit ordering of the pure powers that would make the expansion a series in the jumps, are not proved here. Theorem 4.1 is an identity between two finite sums, and it is stated over the full box rather than the smooth prefix, so it is not a statement about L at a cutoff.

The tail dynamics that such an expansion would feed is recorded in the same module as a one-step map, $\tau(b, d, s) = b(s - d)$, the variable-base tail step, together with the rewriting that names its expanded form. No orbit of this map is analysed.

5 Shell multiplicity

A tail estimate needs to know how many lattice points can share a short multiplicative interval. Write $\mathcal{S}$ for the set of exponent triples in a box $B(h_p, h_q, h_r)$ whose smooth value lies in $[\lambda, \eta)$, the smooth exponent shell.

Lemma 5.1 (uniqueness in a short interval). *Let $b \geq 1$ and suppose $\eta \leq b\lambda$. If $b^a w$ and $b^{a'} w$ both lie in $[\lambda, \eta)$, then $a = a'$.*

Proof. If $a < a'$ then $\eta \leq b\lambda \leq b \cdot b^a w = b^{a+1} w \leq b^{a'} w < \eta$, which is impossible; the case $a > a'$ is symmetric. $\square$

Formalised as the short-interval uniqueness. The hypothesis is that the interval has multiplicative width at most b; then one coordinate is determined by the other two, and projecting it away is injective.

Proposition 5.2. *If $\eta \leq r\lambda$ then $\#\mathcal{S} \leq (h_p+1)(h_q+1)$; if $\eta \leq p\lambda$ then $\#\mathcal{S} \leq (h_q+1)(h_r+1)$.*

Formalised as the third-coordinate projection and the first-coordinate projection.

Theorem 5.3 (quadratic multiplicity bound). *Suppose $\eta \leq r\lambda$ and $h_p \leq h_q \leq h_r$ with $h_p + h_q + h_r = j$. Then*

$$9\,\#\mathcal{S} \leq (j+3)^2.$$

Proof. By Proposition 5.2, $\#\mathcal{S} \leq (h_p + 1)(h_q + 1)$. It therefore suffices to prove that $a \leq b \leq c$ with $a + b + c = j$ gives $9(a+1)(b+1) \leq (j+3)^2$. From $a \leq b \leq c$ we get $a + 2b \leq j$, so it is enough that $9(a+1)(b+1) \leq (a+2b+3)^2$; writing $b = a + d$ with $d \geq 0$, the difference of the two sides is $d(3a + 4d + 3) \geq 0$. $\square$

Formalised as the quadratic shell bound, over the sorted quadratic estimate. The constant 9 is the square of the number of generating primes and appears because the bound is the arithmetic–geometric comparison for a sum of three sorted coordinates; sorting is a hypothesis, not a normalisation, since the shell itself is not symmetric in the three bases.

The bound is uniform in λ and η subject to the width condition, and it is stated for the actual filtered shell rather than for a lattice model of it. It is an input to a tail estimate and is not itself one: no series is bounded here.

6 The kernel does not separate

A recurring hope is to write the three-prime kernel as a product of functions of the separate coordinates, which would reduce the problem to one-dimensional criteria. At the smallest instance this fails, and the failure is exact.

Theorem 6.1 (non-separability at $\{2,3,5\}$). *With* $(p,q,r) = (2,3,5)$*,*

$$\mathrm{K}(0,0,0)\ \mathrm{K}(1,1,0) = \tfrac{1}{60} \neq \tfrac{1}{12} = \mathrm{K}(1,0,0)\ \mathrm{K}(0,1,0).$$

Proof. The four values are $\mathrm{K}(0,0,0) = 1$, $\mathrm{K}(1,0,0) = 1/2$, $\mathrm{K}(0,1,0) = 1/6$ and $\mathrm{K}(1,1,0) = 1/60$, computed from $\mathrm{H}(1) = 1$, $\mathrm{H}(2) = 2$, $\mathrm{H}(3) = 2\cdot 3 = 6$ and $\mathrm{H}(6) = 4\cdot 3\cdot 5 = 60$. □

Formalised as the non-separation witness, over the four exact values, the origin, value at two, value at three, and value at six. The height at 6 is 60 rather than 6 because the maximal pure powers below 6 are 4, 3 and 5: the running least common multiple at a smooth cutoff sees powers of the other primes that the cutoff itself does not contain. That is the mechanism behind the non-separation.

A nonzero two-by-two minor rules out writing the kernel as $f(i)g(j)h(k)$ on this box, and hence rules out any argument that first separates the coordinates and then treats them one at a time. It does not bound the rank of the kernel from below beyond one, and it is not an independence or irrationality statement.

7 The exact local-window frontier

7.1 The actual four-symbol dyadic radix

Compress the jump word between consecutive powers of two: a block starts just after 2^a, includes every pure 3- or 5-power strictly between 2^a and 2^{a+1}, and ends with the jump at 2^{a+1}. A channel cannot occur twice inside one block. Indeed, if

$$2^a < p^e, p^f < 2^{a+1} \qquad (p \geq 2),$$

then the ratio between the interval endpoints is $2 \leq p$, so strict monotonicity of the powers forces $e = f$. This is the checked internal-power uniqueness lemma.

Let β_a be the product of the terminal dyadic factor 2, a factor 3 when the block contains an internal 3-power, and a factor 5 when it contains an internal 5-power.

Theorem 7.1 (the dyadic block alphabet). *For every* a*,*

$$\beta_a \in \{2,6,10,30\}; \qquad \textit{in particular} \qquad 2 \leq \beta_a \leq 30.$$

Proof. Internal-power uniqueness leaves two independent yes/no choices, one for the 3-channel and one for the 5-channel. Multiplying the terminal factor 2 by the selected channel factors gives exactly the displayed four cases. □

The definition is the dyadic block base; Lean checks both the exact four-case alphabet and the bounded-radix consequence. Thus radix growth is no longer an open part of the producer. What is not yet checked in Lean is the identification of the corresponding block digit with the multiplicity forcing in the original series.

Exact finite consumer experiment. The integer-only dyadic-window checker constructs the ordered pure-power jumps, the block bases, and the block digits from exact multiplicity counts. It reproduces the following certificates; the columns are denominator B, dyadic start a, window length h, endpoint jump index n, window base W, forcing F, least positive residue $R = \mathrm{lpr}_W(-BF)$, and short bound K.

B	a	h	n	W	F	R	K
1	1	2	4	60	47	13	9
7	1	3	6	360	289	137	95
16	1	4	9	10800	8735	640	352

A fresh scan over every $B \le 1000$ coprime to 30 and every $100 \le a \le 500$ tested 106,666 pairs. In every case a window of length at most 18 made both $W > K$ and $\mathrm{lpr}_W(-BF) > K$; the largest first successful length was 14. The computation uses integers only and is reproducible from the pinned checker. Its claim ceiling is deliberately finite: neither the scan nor the three displayed certificates proves escape for unbounded B or cofinally many starts.

The residue route can now be stated without hiding either the denominator or the consumer. For an integer radix word b_n and forcing word m_n, define

$$W_{\ell,0} = 1, \quad W_{\ell,h+1} = b_{\ell+h} W_{\ell,h},$$
$$F_{\ell,0} = 0, \quad F_{\ell,h+1} = b_{\ell+h} F_{\ell,h} + m_{\ell+h}.$$

These are the window base and window forcing. If an integral carry satisfies

$$d_{n+1} = b_n d_n - B m_n,$$

then induction gives the exact division-free identity

$$d_{\ell+h} = W_{\ell,h} d_\ell - B F_{\ell,h};$$

this is the checked window identity.

For $C > 0$, let $\mathrm{lpr}_C(x) \in \{1, \dots, C\}$ be the least positive representative of $x \bmod C$, with a zero residue represented by C. This is the canonical representative. Lean checks both its positive range and its congruence to the source integer. This convention matters: replacing $\mathrm{lpr}_C(x)$ by $|x|$ would not be a modular statement.

Let $K(B, n)$ be the bound on the reduced carry at denominator B. Define *cofinal local-window escape* to mean that for every $B > 0$ coprime to 30 and every ℓ_0, there are $\ell \ge \ell_0$ and $h > 0$ such that

$$C_{\ell,h} := |W_{\ell,h}| > 0 \quad \text{and} \quad K(B, \ell + h) < \mathrm{lpr}_{C_{\ell,h}}(-BF_{\ell,h}). \tag{E}$$

The quantifier over B is part of the proposition, and so is the dependence of K on B; suppressing it gives the wrong expert interface. The exact unproved proposition is the cofinal local-window producer.

Theorem 7.2 (the finite residue contradiction)**.** *Let $C > 0$, $c > 0$, and $|c| \le K$. If $c \equiv x \pmod{C}$ and $K < \mathrm{lpr}_C(x)$, then the hypotheses are contradictory.*

Proof. The canonical representative lies in $\{1, \dots, C\}$. The inequalities put c strictly between 0 and C. If $x \equiv 0 \pmod{C}$, its positive representative is C while $c \bmod C = c \neq 0$. Otherwise both c and $\mathrm{lpr}_C(x)$ are their own residues and congruence makes them equal, contradicting $|c| \leq K < \mathrm{lpr}_C(x)$. □

The natural-state version is the finite consumer; the integer carry version is the least-positive-residue consumer.

Theorem 7.3 (the checked local-window consumer)**.** *Assume the cofinal escape property* (E)*. Fix $B > 0$ coprime to 30. There is no integral sequence d_n satisfying simultaneously*

$$d_{n+1} = b_n d_n - Bm_n, \qquad d_n > 0, \qquad |d_n| \leq K(B, n) \quad (n \geq 0).$$

Proof. Choose one window supplied by (E). The checked window identity gives

$$d_{\ell+h} \equiv -BF_{\ell,h} \pmod{|W_{\ell,h}|}.$$

The endpoint state is positive and at most $K(B, \ell + h)$, whereas the canonical positive residue of the right-hand side is larger than this bound. Theorem 7.2 is the contradiction. □

This is formalised as the reduced-carry extinction theorem. Coprimality with 30 is used by the open producer to select a window; once a window has been emitted, the finite contradiction does not use it. The theorem also treats the edge cases explicitly: $|W_{\ell,h}| = 0$ is excluded, a zero residue is represented by the full modulus, and positivity prevents the endpoint carry from being zero.

7.2 Candidate mechanisms and disconfirmers

The most direct mechanism is now four-symbol anti-concentration. The radix word is constrained to $\{2, 6, 10, 30\}$, so the surviving arithmetic lies in the block digit and its carry into the residue class. One seeks a cofinal family of blocks for which the modulus $|W_{\ell,h}|$ grows faster than the carry window and the weighted forcing $-BF_{\ell,h}$ does not remain trapped in the two short arcs adjacent to 0 modulo that modulus. A theorem for one cofinal family is enough; no equidistribution of all windows is required.

Four shortcuts are already disconfirmed. Merely proving $2 \leq b_a \leq 30$ loses the exact channel alphabet. A fixed finite list of denominator certificates does not address the universal B-quantifier. A bound independent of B is not the bound delivered by denominator clearing. Finally, large absolute forcing is irrelevant unless its *least positive residue* is large. These are not stylistic cautions: each removes a false version of (E).

8 Specialist handoff and surviving questions

Problem 8.1 (the exact surviving producer)**.** For the actual ordered $\{2, 3, 5\}$ jump expansion, identify the formal radix word with $\beta_a \in \{2, 6, 10, 30\}$, define and verify the corresponding block digit m_a, derive the denominator-cleared bound $K(B, a)$ from a hypothetical rational value, and prove (E) for every $B > 0$ coprime to 30 and every starting index. Equivalently, prove a cofinal subfamily of windows on which

$$\mathrm{lpr}_{|W_{\ell,h}|}(-BF_{\ell,h}) > K(B, \ell + h).$$

Wanted. A theorem proving Problem 8.1, or a strictly weaker four-symbol block statement that still emits one escaping window beyond every level for each fixed B. The answer must state the block digit, all dependence on B, and the endpoint at which the carry bound is evaluated.

Already checked. Lean checks the exact dyadic radix alphabet, the window recurrence, the least-positive-residue interval and congruence lemmas, the denominator-dependent producer type, and the complete producer-to-contradiction theorem. The structural height, jump, fibre, quadratic shell, and non-separation results earlier in this note are also checked. The exact checker constructs actual block digits and supplies the finite evidence reported above. No Lean theorem yet identifies those digits with the formal forcing word, derives $K(B, a)$ from a rational value, or proves (E).

Most useful answer. First close the exact bridge from the original summands to the block digit used by the checker and to the denominator-cleared bound. Then give an explicit cofinal rule in the alphabet $\{2, 6, 10, 30\}$, together with a lower bound for the canonical residue and an upper bound for $K(B, \ell + h)$. A negative answer is also valuable if it constructs infinitely many (B, ℓ_0) for which every proposed window family fails; that would retire the present producer rather than merely weaken it.

If answered. Theorem 7.3 immediately rules out the corresponding positive reduced carry. After the separate rational-value-to-actual-carry instantiation is checked, this supplies the missing contradiction for the three-prime case. Until both pieces are present, Problem #269 remains open.

Problem 8.2 (direct nonintegral tails)**.** As an alternative to local windows, prove directly that for every positive B coprime to 30 the denominator-cleared actual dyadic tail is nonintegral cofinally. Such a theorem would bypass (E), but it must still feed the same rational-carry consumer.

Problem 8.3 (a genuinely three-dimensional analytic theorem)**.** Prove a target-faithful analytic statement for the phase cocycle attached to the three logarithmic coordinates. By Theorem 6.1, a preliminary separation into one-dimensional factors is unavailable even at $\{2, 3, 5\}$.

Statements and declarations

Artefact and data availability. The pinned formal-source revision contains the Lean sources, the fixed toolchain, the library manifest, and the exact dyadic-window checker used in the finite experiment. This manuscript provides navigation rather than proof authority.

Declaration of generative AI use. Large-language-model agents were used throughout development to draft and revise prose, formal proofs, and software. The author set the objectives and acceptance criteria, selected and reviewed the claims, and approved the published version. The author assumes responsibility for the accuracy, interpretation, and presentation of the work. Generative systems are production tools; they are not authors and supply no independent authority. Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible.

Funding and competing interests. This work received no external funding. The author declares no competing interests.

Acknowledgements. The problem numbering and status follow the Erdős Problems catalogue maintained by Thomas Bloom [4].

A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision 773d2a6ec564. The running-LCM structure, residue arithmetic, and local-window bridge occupy three separate modules. Three distinctions are worth carrying into the source. The height statements hold for arbitrary bases, while the statements about L need the three primes to be distinct. The normal form of Section 4 is a finite identity over a rectangular box, not a convergence theorem. And the formal cofinal-escape predicate is an unproved proposition consumed by a theorem; its application to the actual $\{2, 3, 5\}$ word is not asserted.

References

[1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. 28, Geneva, 1980, p. 65.

[2] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.

[3] P. Erdős, letter to the editor (written 1 January 1973), Fibonacci Quart. **12** (1974), p. 335.

[4] T. F. Bloom, *Erdős Problem #269*, `erdosproblems.com/269`, accessed 22 July 2026.

[5] The Formal Conjectures Authors, *FormalConjectures.ErdosProblems.*`269`, Lean source, 2025, accessed 27 July 2026.